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Duration	4 mins 48 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It is slower than linear search for small lists
b. It requires the list to be sorted ✓
c. It can only be applied to large lists
d. It does not work with negative numbers

Question **2**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By sorting the list first
b. By dividing the list into halves
c. By comparing each element in the list sequentially ✓
d. By using a hash function

Question **3**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the middle of the list
b. In the left sublist
c. In the right sublist ✓
d. At the beginning of the list

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search is more efficient than binary search on large lists.
b. Linear search divides the list into halves.
c. Linear search requires the list to be sorted.
d. Linear search can be applied to both sorted and unsorted lists. ✓

Question **5**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It compares elements sequentially
b. It can be applied only if the list is sorted ✓
c. It always starts from the beginning of the list
d. It works on unsorted lists

Question **6**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Jump search
b. Binary search
c. Hashing
d. Linear search ✓

Question **7**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By using a hash function
b. By starting from the first element
c. By comparing each element sequentially
d. By dividing the list length by two ✓

Question **8**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. n ✓
- b. n/2
- c. n-1
- d. log n

Question **9**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

- a. Binary search
- b. Linear search ✓
- c. Hash search
- d. Both (1) & (3)

Question **10**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- a. 100
- b. 10 ✓
- c. 1000
- d. 500

Question **11**

Incorrect

Mark 0.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. The list must be sorted

b. Number values should only be present ✗

Number values should only be present

c. There must be a mechanism to delete and/or insert elements in the list

d. There should be direct access to the middle element in any sublist

Question **12**

Incorrect

Mark 0.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 90 and 99

b. 89 and 94

c. 94 and 99 ✗

d. 90 and 100

Question **13**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Hash

b. Both (1) & (3)

c. Binary ✓

d. Linear

Question **14**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(n log n)
b. O👎
c. O(log n) ✓
d. O(1)

Question **15**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is small and unsorted ✓
b. When the list is large and unsorted
c. When the list is small and sorted
d. When the list is large and sorted